Rhythms

Introduction (Part 1)

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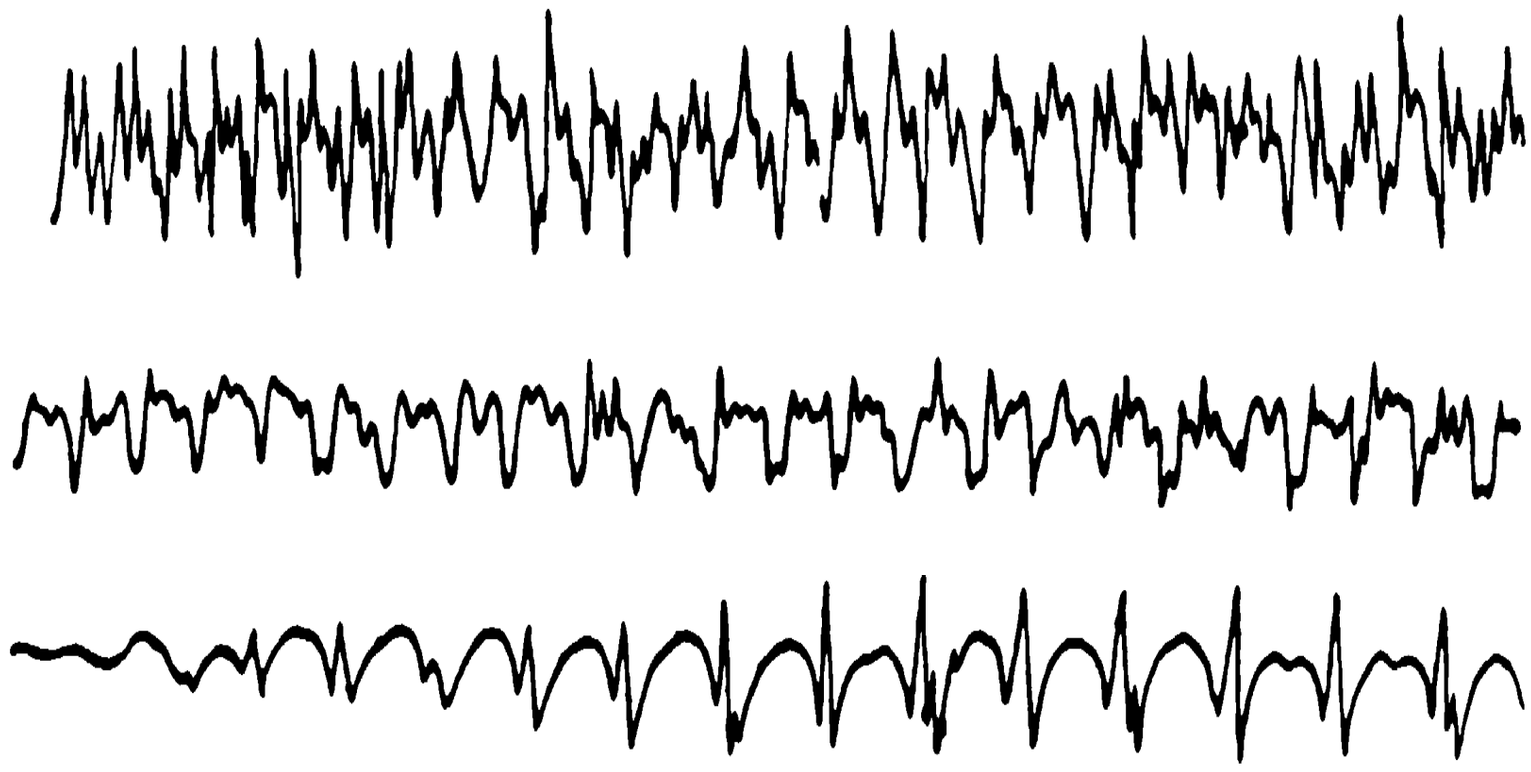
Brain rhythms

Fact: The brain can generate rhythmic activity.

Ex: Scalp electroencephalogram (EEG)

Note: Rhythms also appear in LFP, MEG, fMRI, ...





Observe: Different shapes. Different frequencies.

Brain rhythms: characterization

Q: How do we characterize these rhythms?

To start, we can visualize and describe the rhythms.

Typical features:

Amplitude: Large or small ?

Frequency: Fast or slow ?

Shape: Sinusoidal, square, triangle, . . . ?

Duration: Long or short lasting ?

Our focus (usually) is frequency: How fast or slow is a rhythm?

Q: Why? Because different frequency rhythms are associated with different functions … (See Week 1 lecture)

Characterize brain rhythms

Many sophisticated tools to do so.

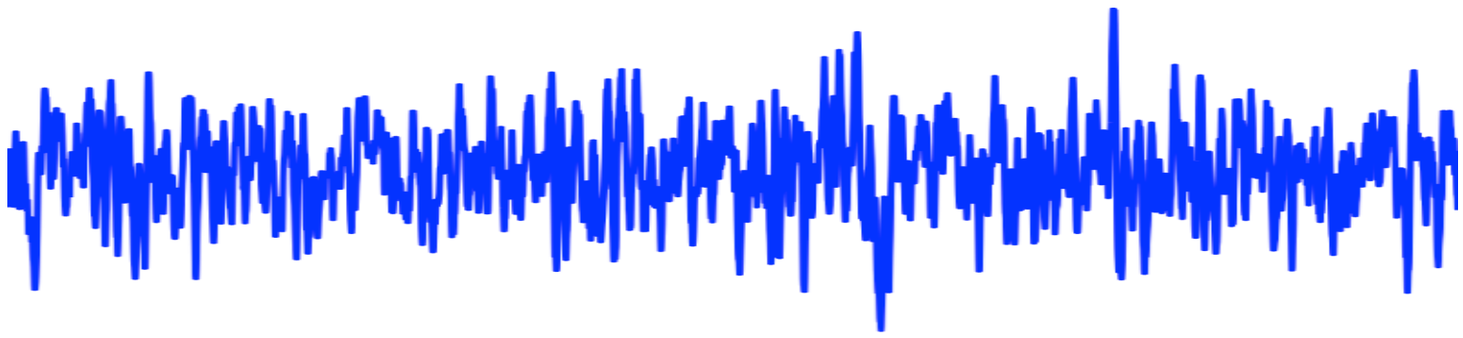
Today, consider two:

–Visual inspection

–Intuition for the spectrum

Idea:

Visual inspection: Plot the data. What do you see?



Spectrum: Break down the data into sinusoids . . .

Remember: sinusoids . . .

$$V[t] = A\cos(2\pi f t)$$

Voltage as a function of time

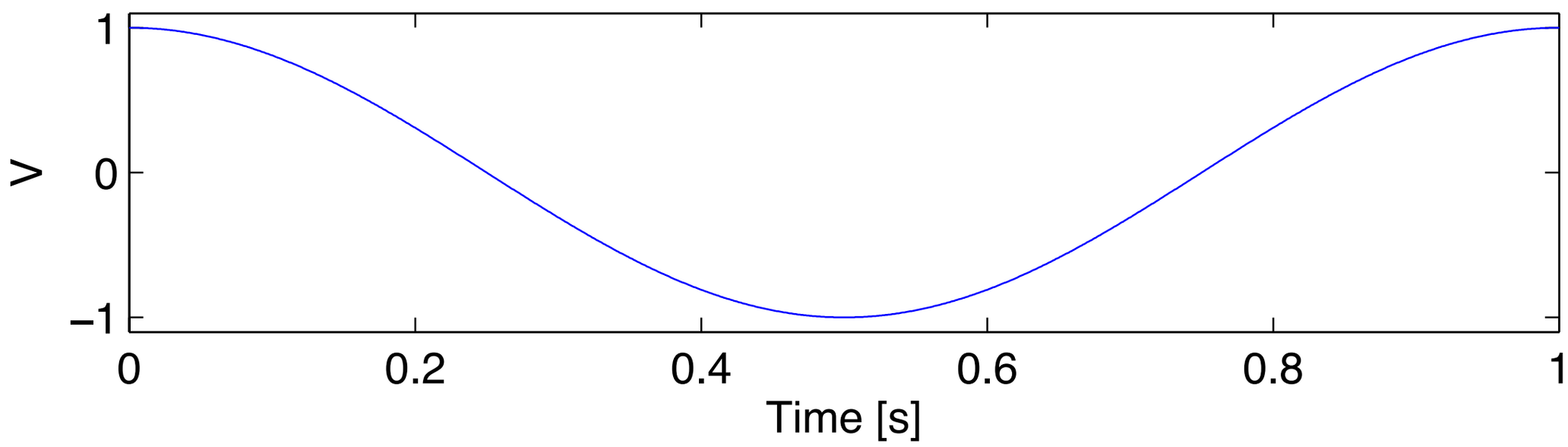
Amplitude

Frequency [Hz]

Time [s]

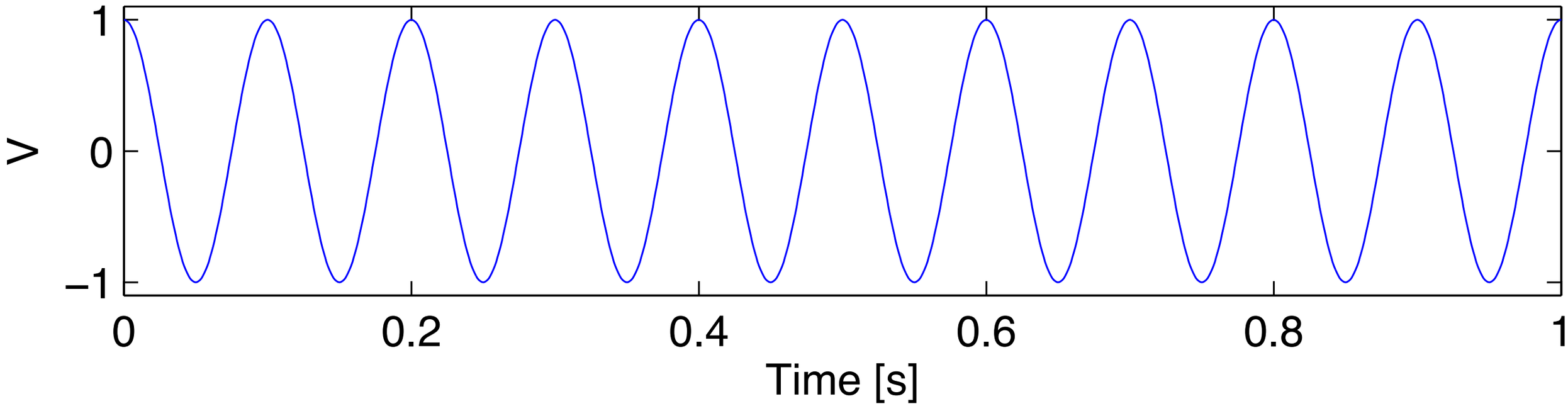
Ex. Consider: f = 1 Hz

Q: What does it look like?



Remember: sinusoids . . .

Q: What is the frequency of this sinusoid?



A: 10 cycles in 1 second, so 1 Hz.

Note: Visual inspection is often useful.

Remember: sinusoids . . .

In addition to cosine, there's also sine:

$$V[t] = B\sin(2\pi f t)$$

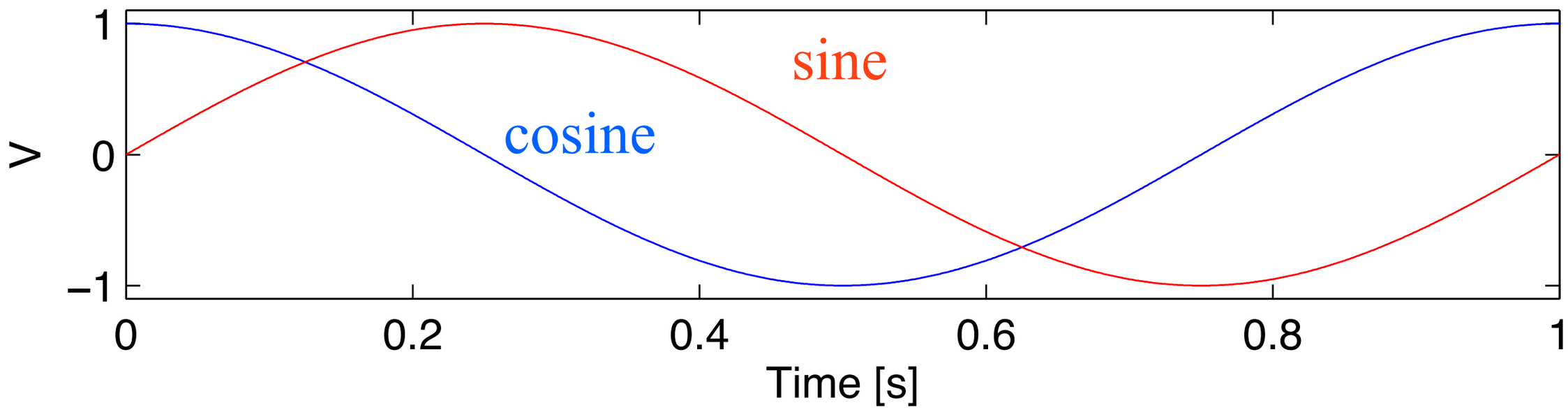
Voltage as a function of time

Amplitude

Frequency [Hz]

Time [s]

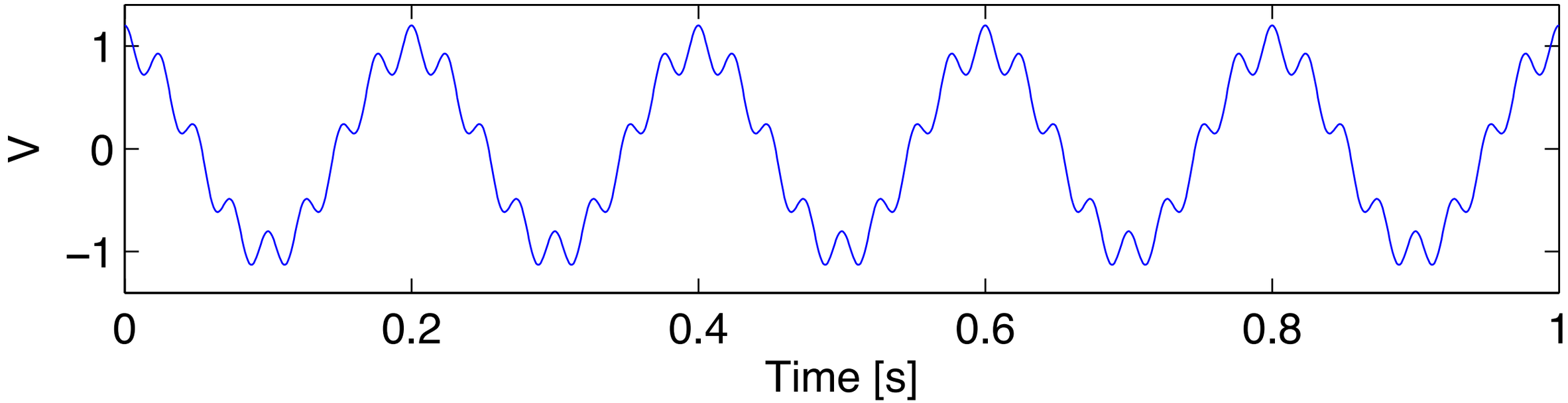
Ex. Consider: f = 1 Hz



Q: What's the difference?

Example: rhythmic signal

Consider the signal below:



Q: What are the rhythms?

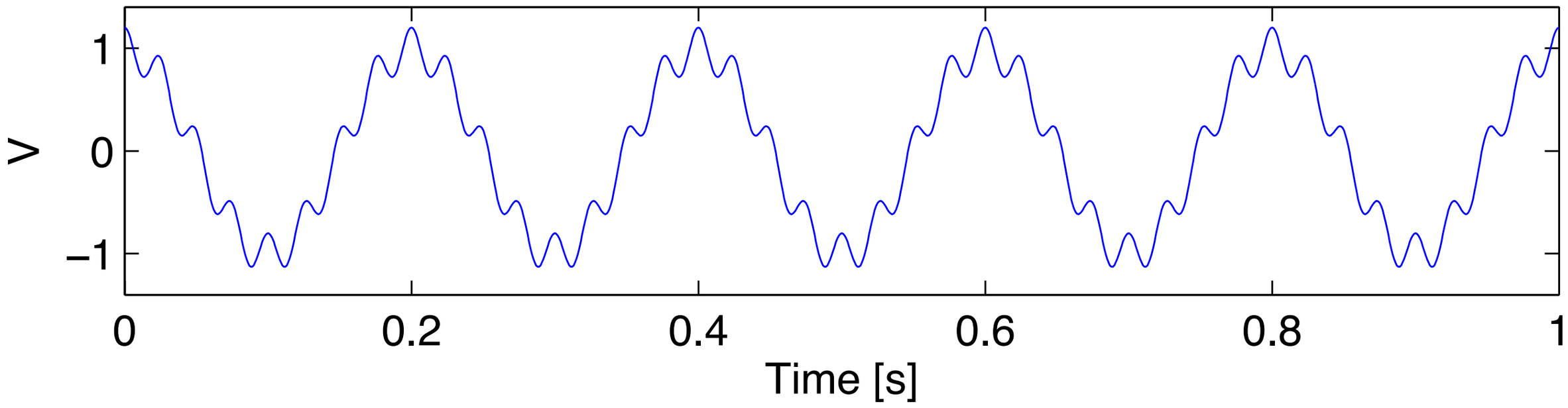
A: Apply visual inspection . . . Slow and fast

5 Hz 40 Hz

Q: What has larger amplitude?

Example: rhythmic signal

So, we can represent this signal . . .



. . . as the sum of two sinusoids:

A slow, large amplitude sinusoid + a fast, small amplitude sinusoid

$$V[t] = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t)$$

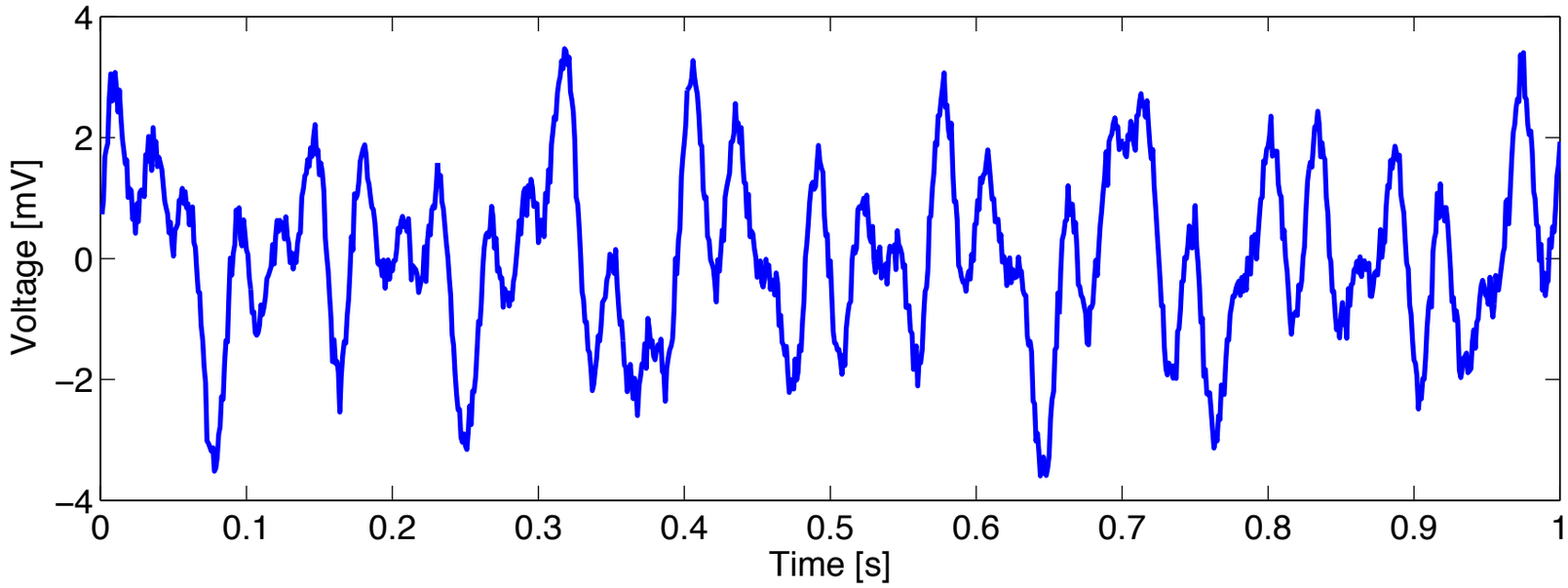
1.0 5 Hz 0.1 40 Hz

We get a <u>simpler</u> representation of the signal. That's the idea of the spectrum.

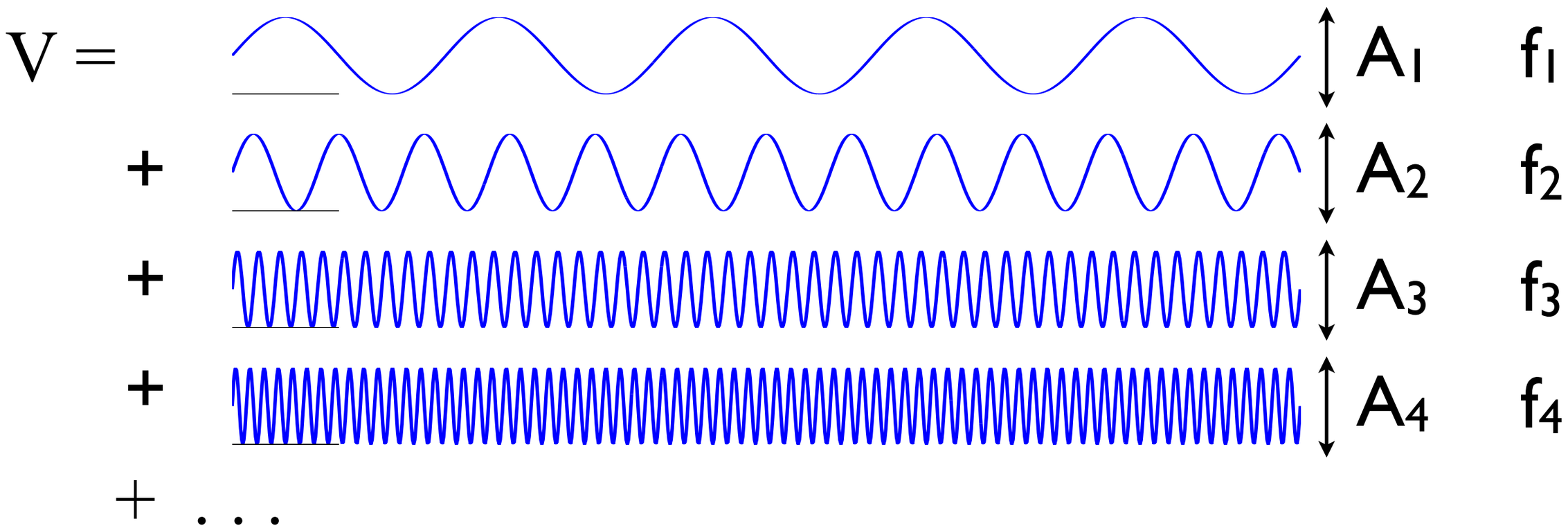
Idea: Spectrum

Consider:

V =



- Decompose signal into oscillations at different frequencies.



Represent V as a sum of sinusoids (e.g., part 7 Hz, part 10 Hz, . . .)

Idea: Spectrum

So, in equations:

$$V[t] = A_1 \cos(2\pi f_1 t) + B_1 \sin(2\pi f_1 t) + A_2 \cos(2\pi f_2 t) + B_2 \sin(2\pi f_2 t) + \ldots$$

amplitude — frequency — amplitude — frequency

Or, more generally:

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

sum over many frequencies — amplitude — oscillation at frequency f_j

Note: A_j and B_j can be zero.
(some rhythms make no contribution to V[t])

Note: A_j and B_j are large when f_j is a good match to the data.

Idea: Spectrum

Note: Think of sine & cosine as accounting for **phase**.

$$C_j \cos(2\pi f_j t + \phi_j)$$

amplitude freq phase

Aside: $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$

$$= C_j \cos(2\pi f_j t)\cos(\phi_j) - C_j \sin(2\pi f_j t)\sin(\phi_j)$$

A_j B_j

$$= A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

Decompose V[t] into sine/cosine or amplitude/phase.

Idea: Spectrum

Q: How do we find A_j and B_j ?

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

A: Consider A_j and use **orthogonality of sinusoids**.

$$\int_0^T \cos(2\pi f_j t)\, \cos(2\pi f_k t)\, dt = \begin{cases} 0 & \text{if} \quad f_j \neq f_k \\ T/2 & \text{if} \quad f_j = f_k \end{cases}$$

Choose T so f_j and f_k complete an integer number of cycles.

frequencies

integrate over time

Idea: Spectrum

Python

Idea: Spectrum

Q: How do we find A_j and B_j ?

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

A: Consider A_j and use **orthogonality of sinusoids**.

$$\int_0^T \cos(2\pi f_j t)\ \sin(2\pi f_k t)\, dt = \quad 0 \qquad \text{for all} \quad f_j, f_k$$



sine

Idea: Spectrum

Return to our original equation

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t) \qquad \underline{\text{Task}}\text{: find } A_j, B_j$$

Pick frequency f_k , multiply both sides by $\cos(2\pi f_k t)$, and integrate over time …

$$\int_0^T V[t]\cos(2\pi f_k t)dt = \int_0^T \sum_j A_j \cos(2\pi f_j t)\cos(2\pi f_k t)dt$$

$$+ \int_0^T \sum_j B_j \sin(2\pi f_j t)\cos(2\pi f_k t)dt$$

Consider each integral …

Idea: Spectrum

by orthogonality

$$\int_0^T \sum_j A_j \cos(2\pi f_j t)\cos(2\pi f_k t)dt \quad = 0 \text{ if } f_j \neq f_k \text{ , or T/2 if } f_j = f_k$$

$$\int_0^T \sum_j B_j \sin(2\pi f_j t)\cos(2\pi f_k t)dt \quad = 0$$

$$A_k\, T/2 \text{ if } j = k, \text{ 0 otherwise}$$

$$\text{So} \quad \int_0^T V[t]\cos(2\pi f_k t)dt = \int_0^T \sum_j A_j \cos(2\pi f_j t)\cos(2\pi f_k t)dt$$

$$+ \int_0^T \sum_j B_j \sin(2\pi f_j t)\cos(2\pi f_k t)dt$$

0

$$= A_k\, T/2$$

Idea: Spectrum

$$\int_0^T V[t]\cos(2\pi f_k t)dt = A_k T/2$$

Solve for A_k

$$A_k = \frac{2}{T}\int_0^T V[t]\cos(2\pi f_k t)dt$$

We've solved for amplitude A_k

Depends on observed data V[t] multiplied by cosine we choose (f_k)

Idea: Spectrum

Return to our original equation

$$V[t] = \sum_j A_j \cos(2\pi f_j t) + B_j \sin(2\pi f_j t)$$

Q: How do we find A_j and B_j ?

A: Consider A_j and use **orthogonality of sinusoids**.

$$A_k = \frac{2}{T}\int_0^T V[t]\cos(2\pi f_k t)dt$$

Similarly

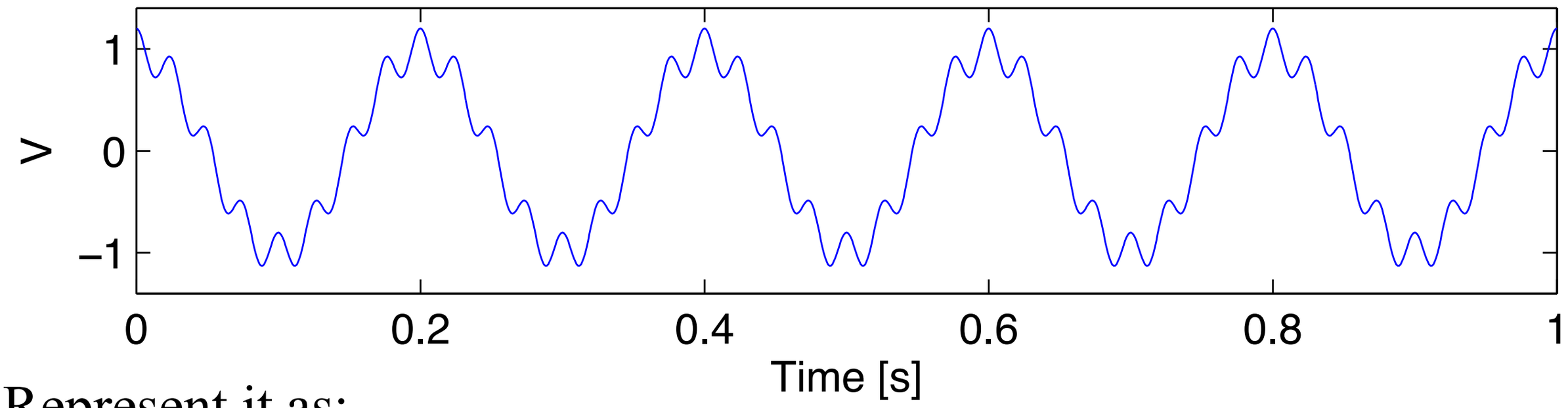
$$B_k = \frac{2}{T}\int_0^T V[t]\sin(2\pi f_k t)dt$$

Big idea: We can decompose V[t] into a sum of sin/cos functions, and we have a strategy to find the amplitudes A_j , B_j

Example: rhythmic signal

Q: So what?

A: Represent V[t] in a simpler way … remember:



Represent it as:

$$V[t] = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t)$$

1.0 5 Hz 0.1 40 Hz

To represent V[t] we need 4 numbers:

Amplitudes = {1, 0.1}

Frequencies = {5, 40} Hz

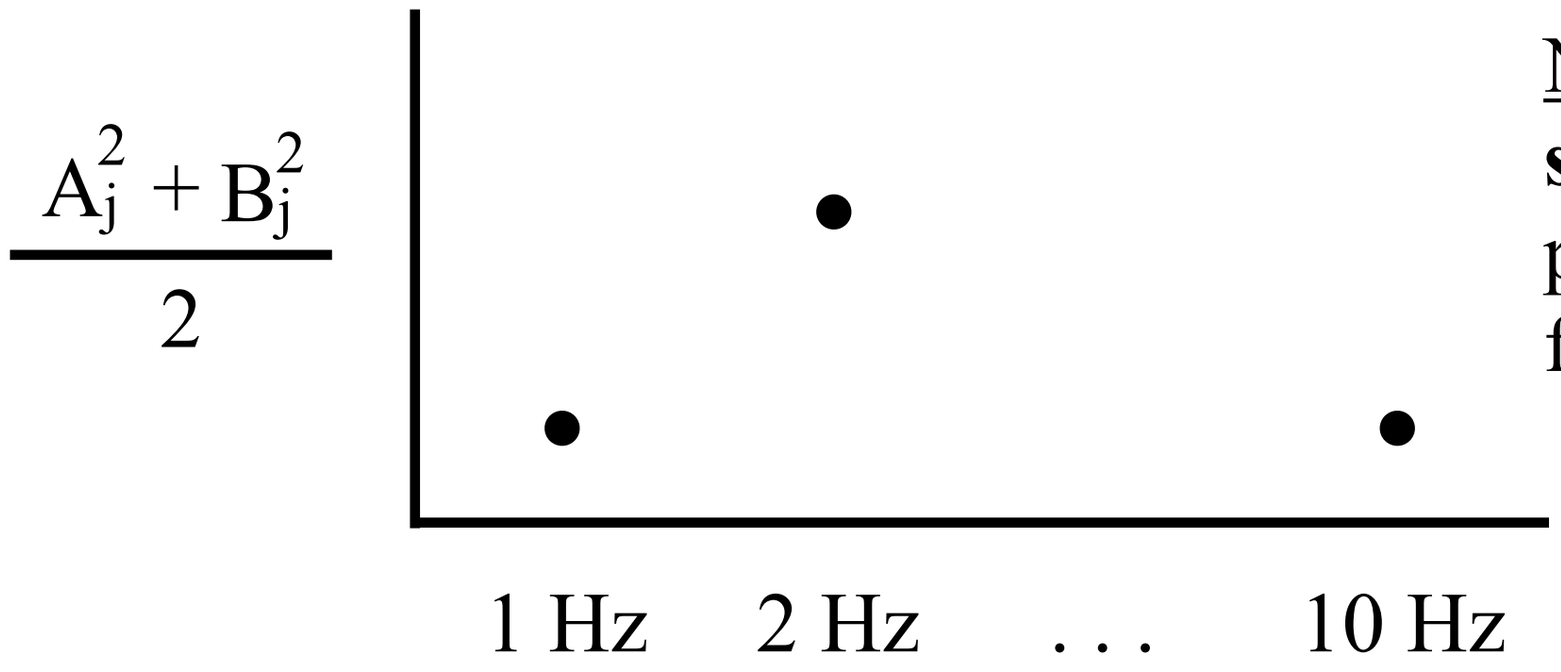
These 4 numbers completely summarize the data.

Plot: Spectrum

We can represent these amplitudes and frequencies graphically:

Plot: $\frac{A_j^2 + B_j^2}{2}$ versus f_j for each j

Note: The summed amplitudes squared. Called the "power"

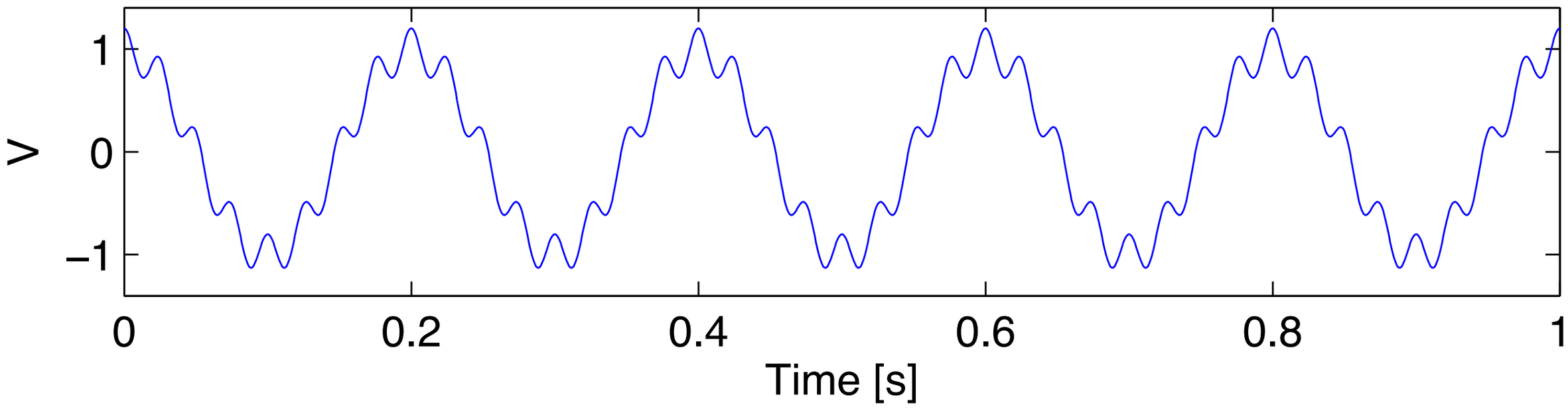


Note: This is called the **spectrum**; a plot of power versus frequency

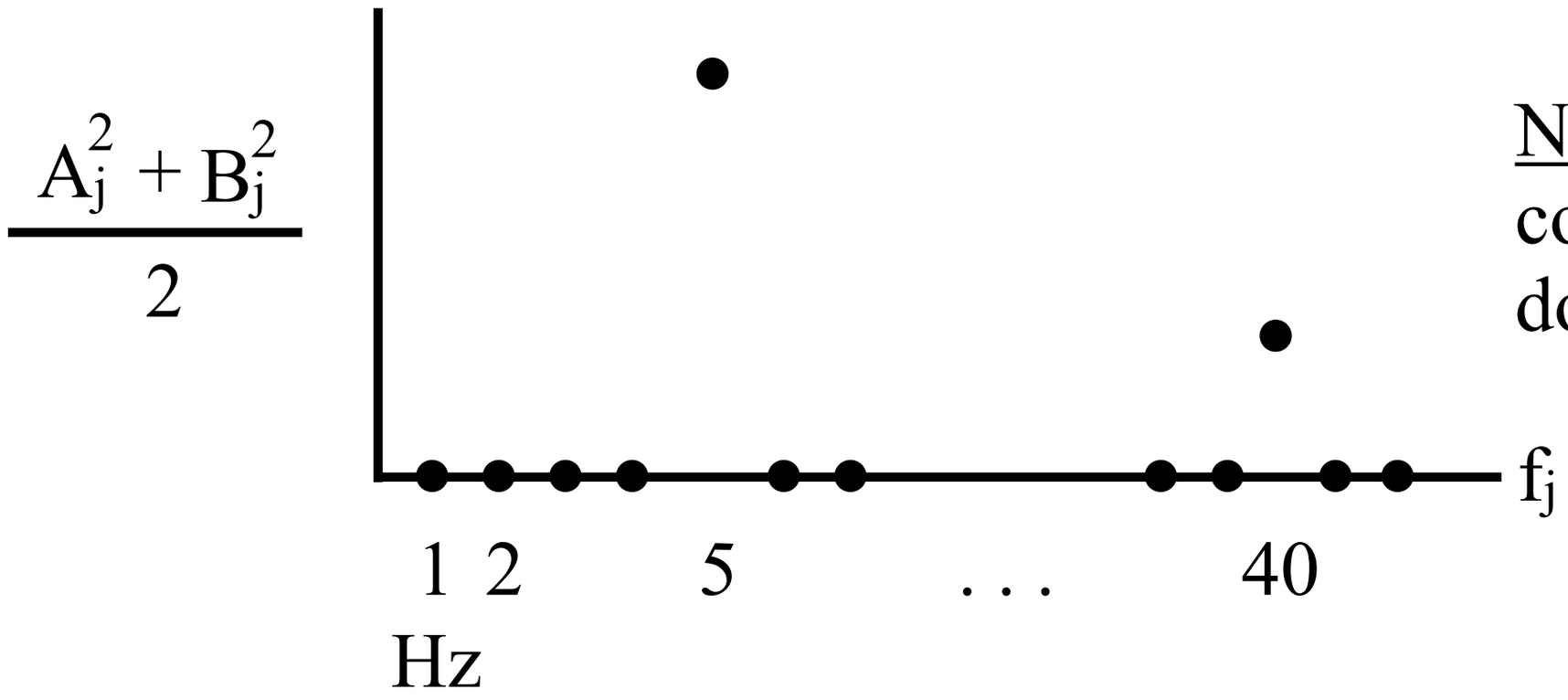
The peaks represent the dominant rhythms in the signal.

Example: rhythmic signal

Ex:



Plot the spectrum:

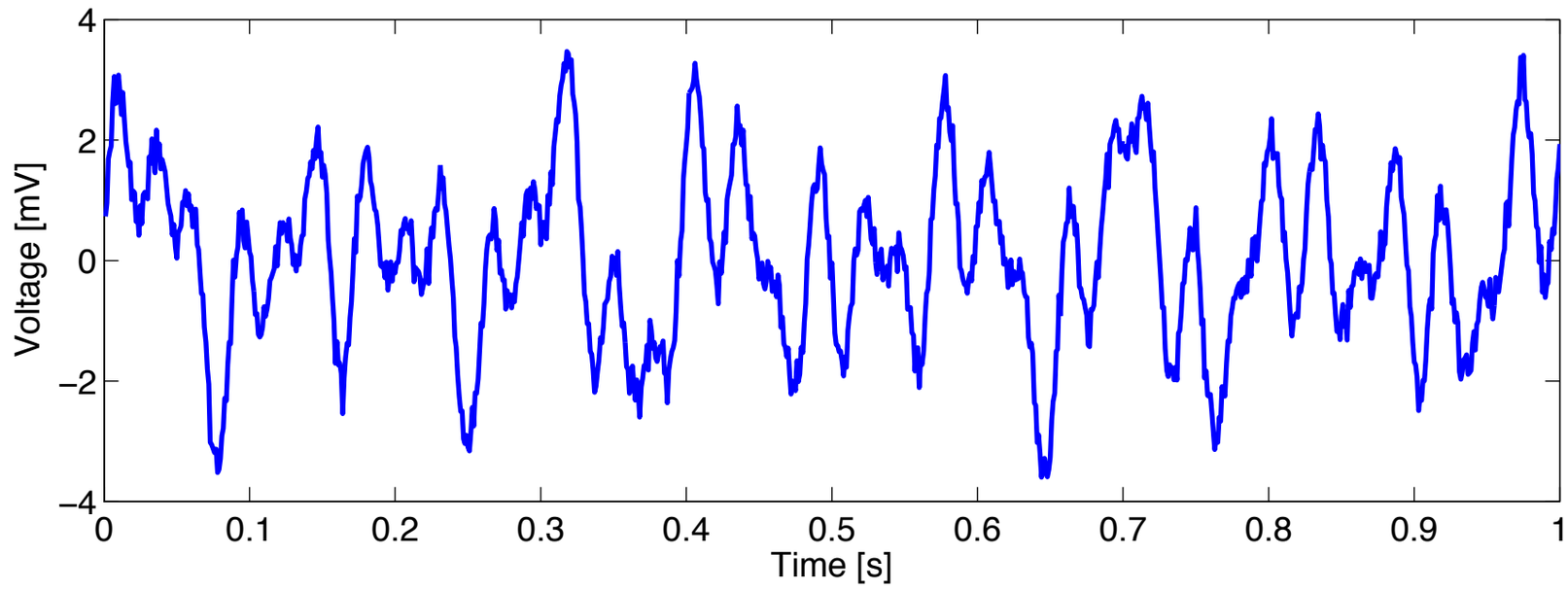


Note: The peaks correspond to the dominant rhythms

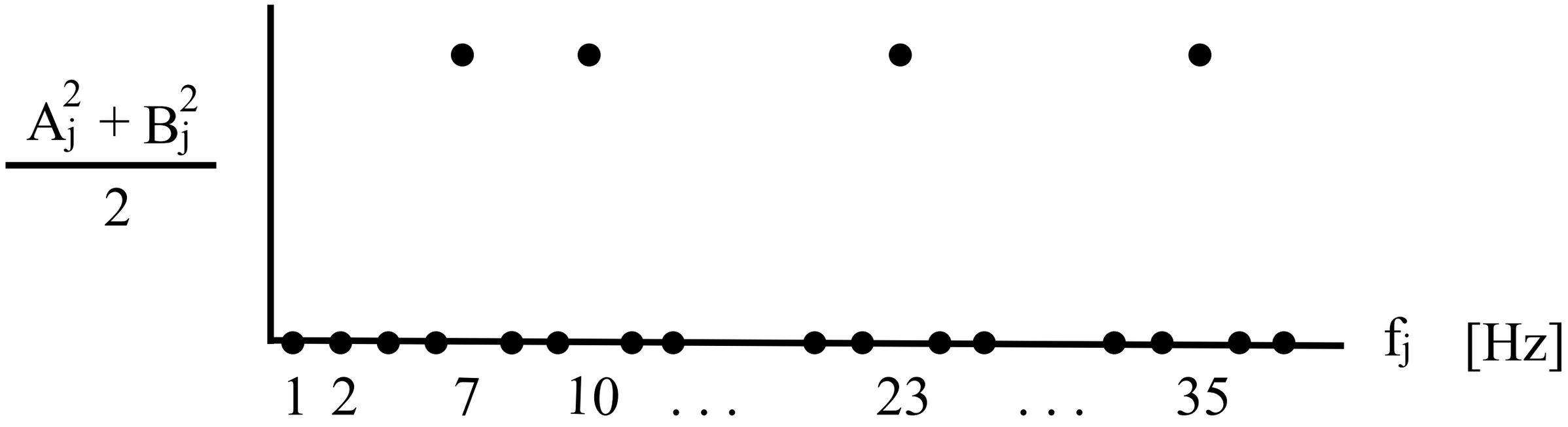
Example: rhythmic signal

Ex.

$V =$



Note: It's complicated Plot the spectrum:

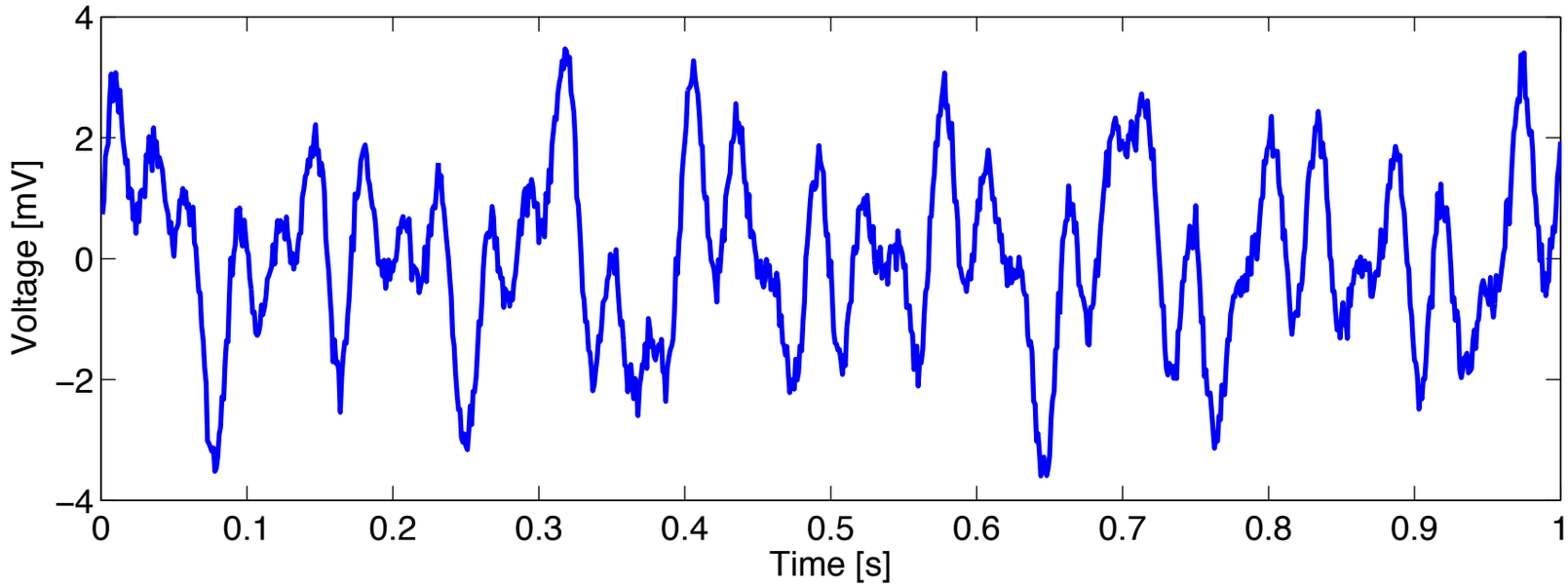


Q: What's happening here?

Example: rhythmic signal

So, by computing the power spectrum, we find the complicated signal:

V =



We find it's the sum of 4 sinusoids at frequencies:

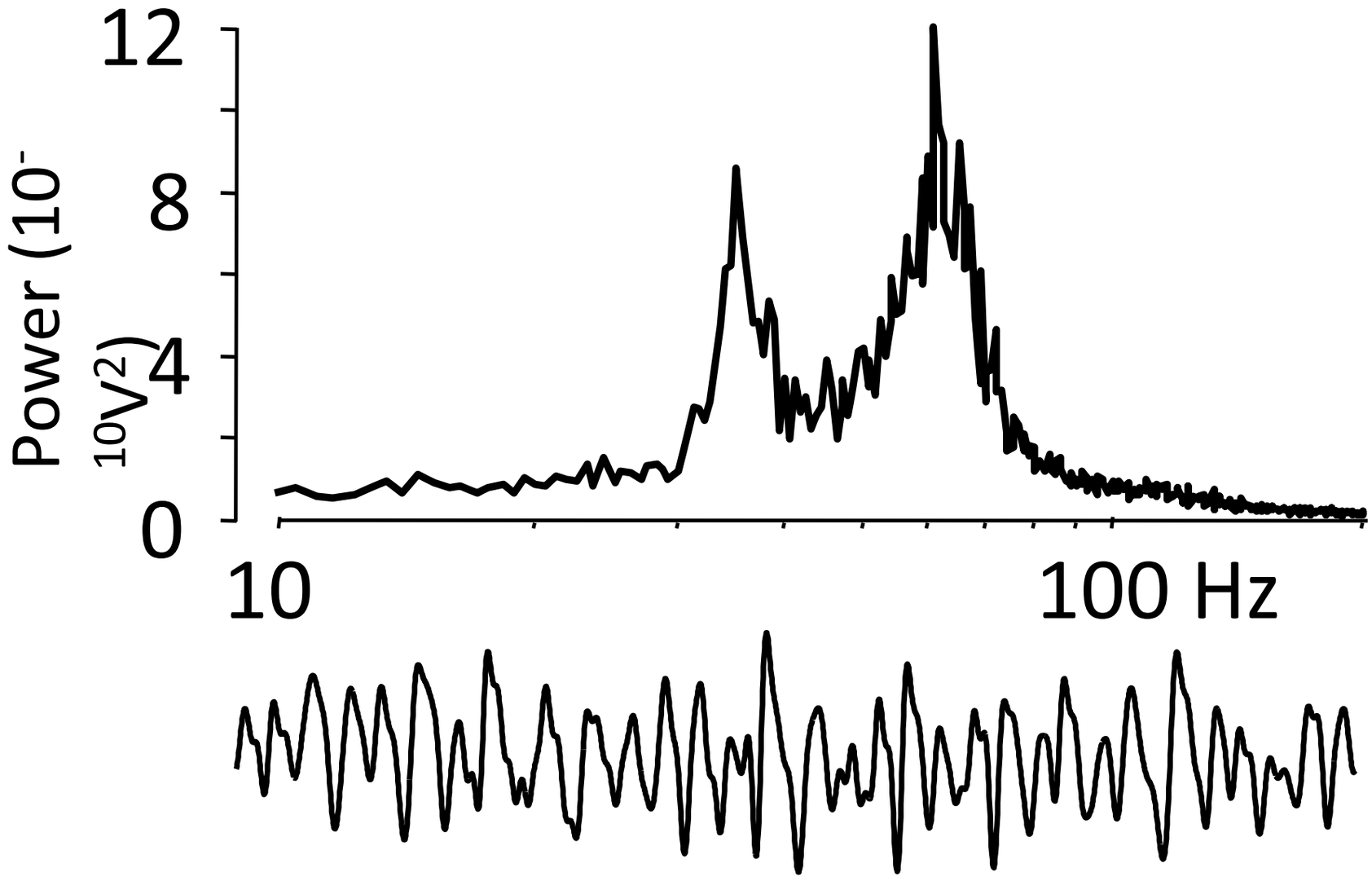
7 Hz, 10 Hz, 23 Hz, and 35 Hz

A much simpler representation of brain activity.

Example: Real world

Q: What does the power spectrum of real-world brain signals look like?

Ex. From a slice of rat cortex:



[Ainsworth et al, Neuron, 2012]

Q: What rhythms are dominant?

Next

Python

Practical notions …